



GASP-NE interim report on UO PM2.5 data in Newcastle City Centre

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1 Summary

The Urban Observatory (UO) has a collection of precision MONITOR sensors. For use in a multi-sensor spatial model as planned for our proposed predictive online tool for the monitoring and forecast of air quality variables (pollutant concentrations). We require as many sensors as possible and therefore verification that the UO MONITOR sensor data agree with that from DEFRA and AURN sensors is needed.

For the purposes of this report we will use the following acronyms:

- 2 DEFRA-owned and approved AURN sensors (DEFRA)
- 5 Locally managed automatic monitoring sensors (Local)
- 5 UO precision MONITOR sensors (UO-Mon)
- UO Mesh sensors (UO-Mesh)

We have highlighted the following as areas for priority action/attention:

- To validate the UO-Mon sensors we compared them to their closest DEFRA/Local sensors; all UO-Mon sensors compared had very good agreement with their closest DEFRA/Local sensor.
- Some UO-Mon sensors were compared with further (in distance) DEFRA/Local sensors. We found an increase in distance to change the relationship between sensors, as expected.
- We will proceed with using the UO-Mon sensors from the Urban Observatory.
- DEFRA Newcastle Centre sensor is systematically rounding PM2.5 values to nearest integer.

The report is structured as follows:

Section 2 presents the locations of all DEFRA, Local and UO-Mon sensors around the Newcastle upon Tyne area that are used within this report on air quality variables (PM2.5, NOx and NO2 concentrations).

Section 3 presents comparisons of PM2.5 concentrations from closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors and closest DEFRA/Local sensors to UO-Mon sensors. March 2026 reasonably demonstrates a longer time period where these sensors could differ in 15min - 1hr intervalled collection. We chose a month that was relatively recent with minimal missing sensors and values. Caveat: there is a short time period on the 21st-22nd of March 2026 where data is missing from all Local sensors.

Section 4 presents a discussion of comparisons of PM2.5 concentrations to NOx and NO2 concentrations from same sensors for DEFRA, Local and UO-Mon sensors. We chose a shorter time scale for this analysis, 30th March - 5th April 2026, as NOx and NO2 concentrations degrade very fast, so only a short time scale is appropriate for their analysis.

Note:

- We have further analysis on wind speed/direction, temperature and altitude relationships with PM2.5 that is not included in this report. We also have comparisons with UO-Mesh sensors.
- When examining wind effects, comparing speed and direction to PM2.5 concentration in 2026 was not possible. With 2025 data, we take old, decommissioned sensors and perform similar analysis with closest UO-Mon sensors and Local sensors. DEFRA sensors were not considered as they are situated further away.

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2 Sensor locations

The three maps in Figure 1 show the closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors. These pairs will be used to analyse correlations between closest sensors and assess the reliability and accuracy of the UO-Mon sensors. The distances (km) between these pairs are stated in Section 3 on the correlation analysis.

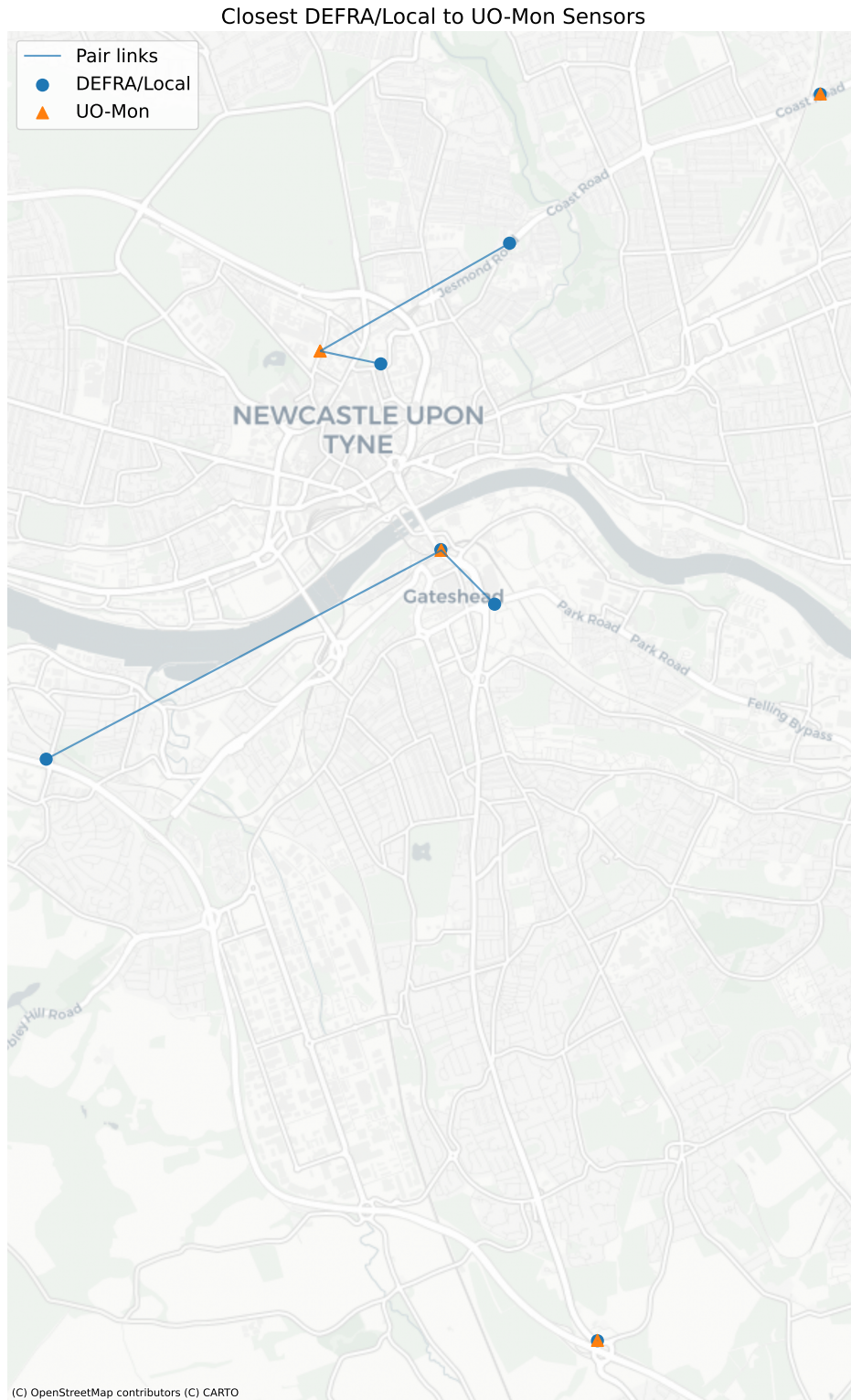


Figure 1: Map of Newcastle upon Tyne area showing locations of the DEFRA/Local sensors closest to the UO-Mon sensors used in the report analysis.

3 Nearby sensor comparisons

We used the sensor data from March 2026 to compare the performance of the UO-Mon sensors with the DEFRA/Local sensors closest to them. March was chosen because this dataset is reliable and consistent with minimal missing values. However, there was still a portion of the data missing around the 21th-22nd from the Local sensors.

Figure 2 presents the data in the form of scatter plots and time series, where only the closest UO-Mon sensor is included for each DEFRA/Local sensor. The aim of this exercise is to assess the quality and accuracy of the UO-Mon sensor data. Both representations of the data demonstrate a very good agreement between the sensors. Figure 3 shows a similar comparison, but for more widely separated sensors. As might be expected we observe a worse agreement. A summary table of correlational information and distances between sensors is given in Table 1 from which we see high correlation between all closest sensor pairs. We use Pearson's correlation coefficients which is sensitive to linear relationships and proportional change between variables. Figure 4 also demonstrates the relationship between distance and the correlation coefficients, and lower correlations are observed when the sensor separation is larger, as expected.

Other relationship (nonlinear) might also exist; we have examined other external factors (such as temperature, wind effects, etc.) that are not included in this report but are available if they are of interest.

In Figure 3, we also examined the agreement between Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge sensors since we have a comparison with UO-Mon Gateshead Tyne Bridge (in the same location). This way we can deduce if the disagreement is indeed related to distance rather than any problems with the sensor quality and accuracy. We see a similar nonlinear relationship occurring between these two sensors as in the case of the Local-Gateshead A1 Dunston and UO-Mon Gateshead Tyne Bridge sensors, which indicates this nonlinear relationship is mainly due to the distance between the sensors.

As shown in Table 1 and Figure 4, the data from the DEFRA and UO-Mon sensors are very consistent, with the correlation coefficients exceeding 0.7 even when the separation of the sensors is as large as 2-3 km. This also suggests the presence of a background PM2.5 distribution which varies little over the distance of a few kilometers. This background and its spatial and temporal variations can thus be identified using the data available.

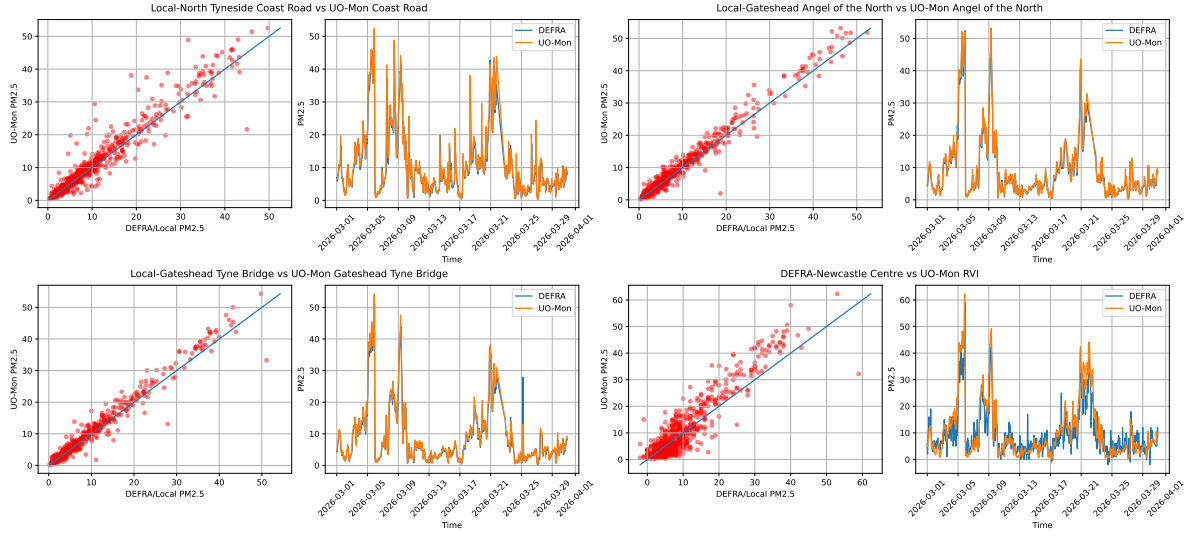


Figure 2: Comparison of PM_{2.5} concentrations measured by each DEFRA/Local sensor and its closest UO-Mon sensor in March 2026. Only the closest UO-Mon sensor is included for each DEFRA/Local sensor. We see very good agreement between the sensors.

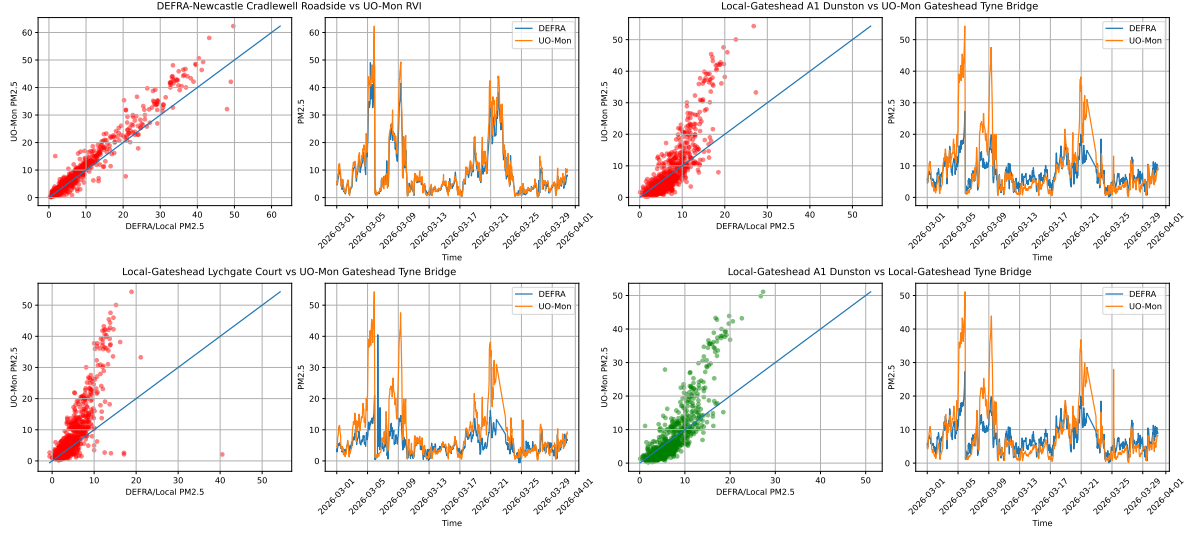


Figure 3: Comparison of PM_{2.5} concentrations measured by DEFRA/Local sensors and additional nearby UO-Mon sensors in March 2026. These comparisons exclude the nearest UO-Mon sensor shown in Figure 2 and illustrate relationships with other, more distant UO-Mon sensors. We see worse agreement with some nonlinear relationships which is expected due to the distance. The lower-right plot (green scatter) shows PM_{2.5} concentrations measured by Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge sensors. We expect to see a similar nonlinear relationship to that comparing Local-Gateshead A1 Dunston and UO-Mon Gateshead Tyne Bridge.

Table 1: Distances between sensors in km, Pearson’s for UO-Mon and closest DEFRA/Local sensors PM2.5 concentrations over long time interval. High correlational coefficients for all close sensors.

DEFRA/Local	UO-Mon	Distance km	Pearsons coeff
Local-Gateshead Angel of the North	UO-Mon Angel of the North	0.003	0.986
Local-Gateshead Tyne Bridge	UO-Mon Gateshead Tyne Bridge	0.006	0.983
DEFRA-Newcastle Cradlewell Roadside	UO-Mon RVI	1.643	0.974
Local-North Tyneside Coast Road	UO-Mon Coast Road	0.002	0.961
DEFRA-Newcastle Centre	UO-Mon RVI	0.468	0.926
Local-Gateshead A1 Dunston	UO-Mon Gateshead Tyne Bridge	3.37	0.846
Local-Gateshead Lychgate Court	UO-Mon Gateshead Tyne Bridge	0.572	0.741

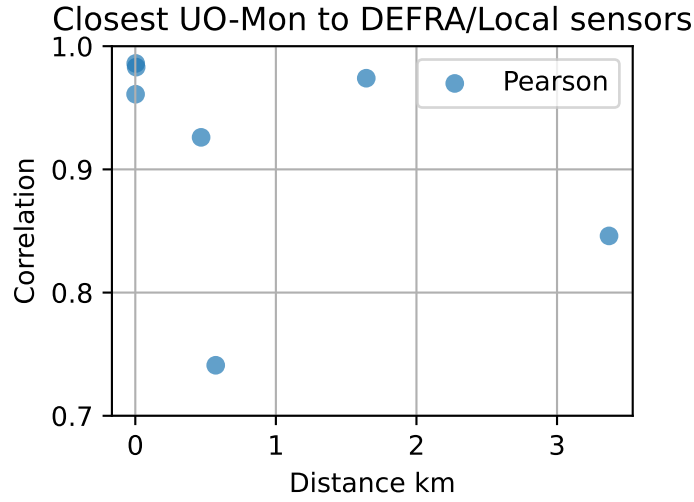


Figure 4: Correlation coefficients of each sensor pair from Figure 2 and Table 1 versus their separation in kilometers.

4 NOx and NO2 comparisons

We compare the PM2.5 and NOx/NO2 concentrations in the same sensors for all DEFRA, Local and UO-Mon sensors using the data from March 30 to April 5, 2026. We chose nitric emissions as a proxy of traffic influence to check whether this has an effect on PM2.5. A shorter time frame was chosen because nitric contamination decays with time occurring much faster than that of PM2.5.

4.1 DEFRA sensors

Figure 5 presents the scatter plots of the PM2.5 and NOx and No2 concentrations measured with the DEFRA sensors, and the legends show the Pearson correlation coefficient. It is clear that the two variable are not well correlated and thus are likely to depend on different factors.

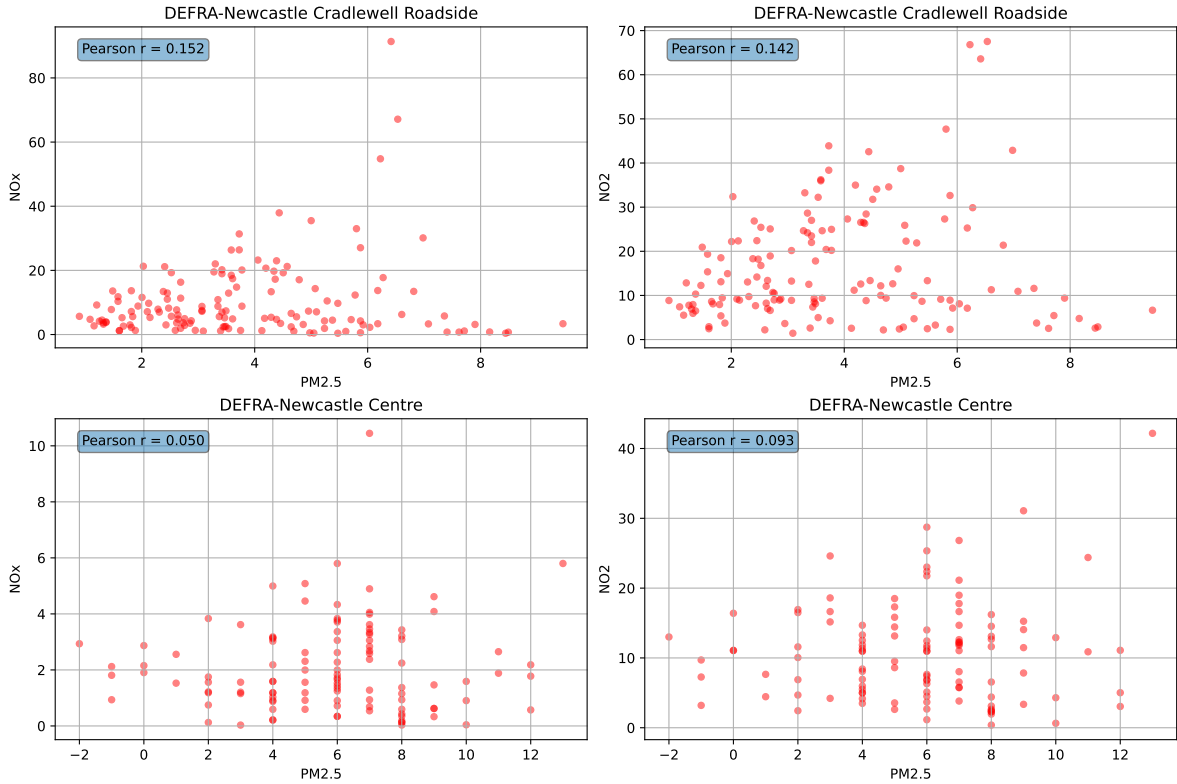


Figure 5: Comparison of PM2.5 and NOx and NO2 concentrations for DEFRA sensors over short time interval (a week).

4.2 Local sensors

Figure 6 presents in a similar manner NO₂ and PM_{2.5} data for the Local sensors (which do not collect NO_x data). The correlation coefficient is rather low and moderate at most. This suggest that, at the location of those sensors, there may be some traffic affecting PM_{2.5} concentration. It cannot be excluded, however, that both NO₂ and PM_{2.5} depend on other factors, such as the air temperature rather than being causally related.

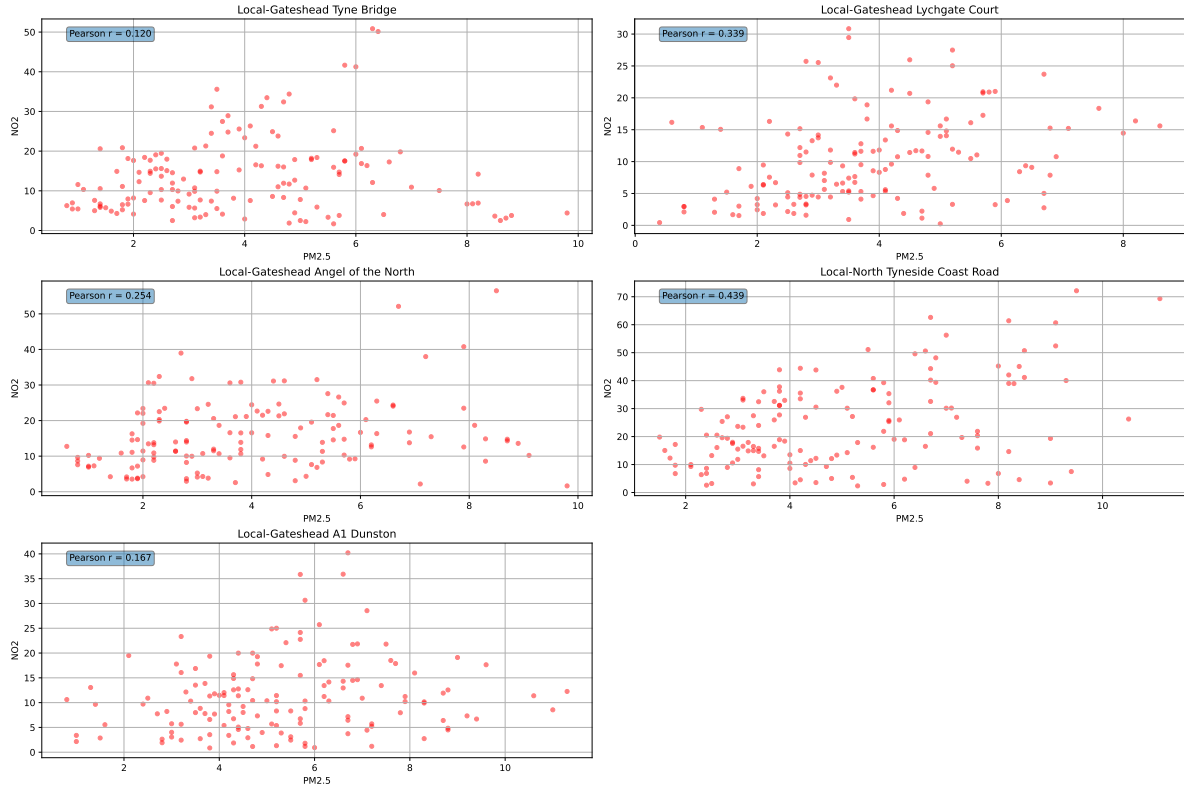


Figure 6: Comparisons of Local sensor PM_{2.5} and NO₂ concentrations over a short time interval (a week).

4.3 UO-Mon sensors

Figure 7 presents similar data for the UO-Mon sensors. As for the other sensors, the correlation is moderate at most and varies strongly between the sensors.

We conclude that the nitric and particulate air contaminations are poorly correlated and one cannot be used as a proxy for the other. In particular, any machine learning model used to forecast one of those variables would not benefit from the input of the measurements of the other one in the training set.

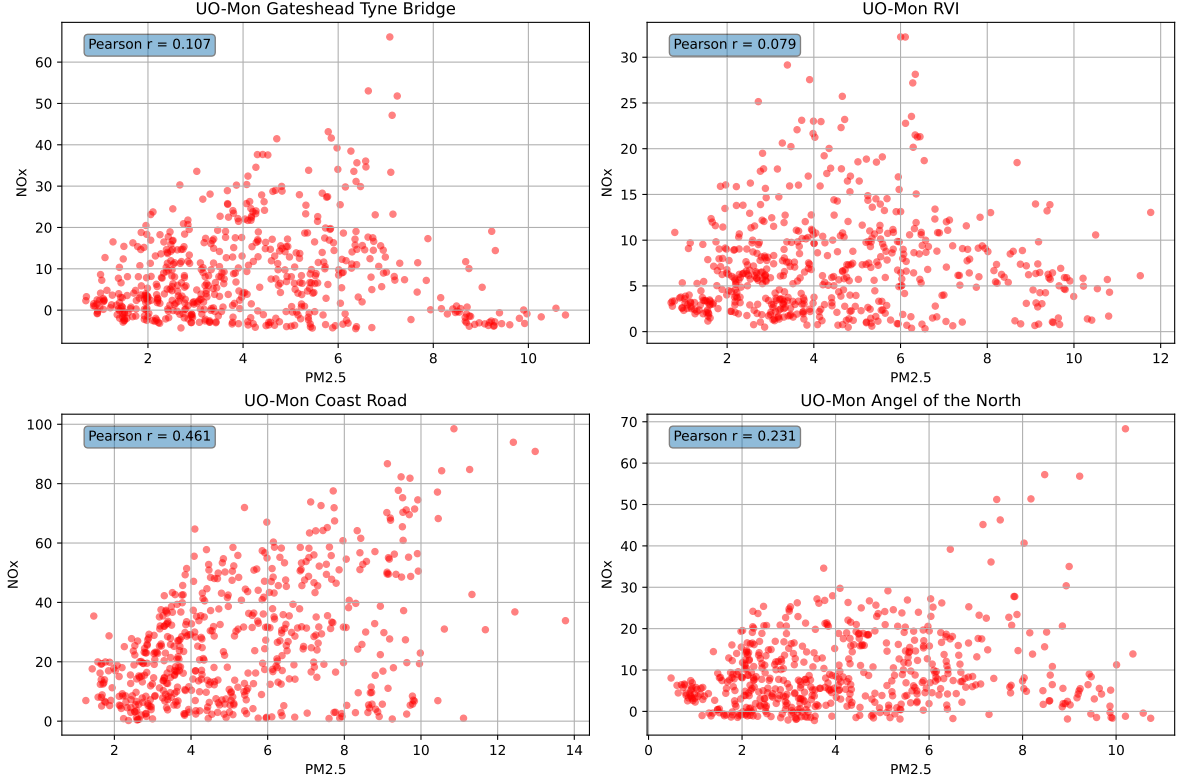


Figure 7: Comparisons of UO-Mon sensor PM2.5 and NOx concentrations over a short time interval (a week).

5 Conclusions

The analyses presented in this report demonstrates that the DEFRA, Local and UO-Mon sensors exhibit sufficient agreement to support their combined use in further modelling. We investigated the influence of external factors, including wind conditions, temperature and sensor altitude, to provide further context for the PM2.5 measurements and inform a predictive machine learning model. Based on this, we suggest to proceed using the DEFRA, Local and UO-Mon sensors to develop a multi-sensor spatial Gaussian Process (GP) model, which will form the basis of the proposed predictive online tool. The inclusion of the UO-Mon sensors would significantly extend the data base for the air quality forecast and allow us to achieve higher reliability and accuracy of the predictions.